



Mediacrest Training College

Nairobi, Kenya

Advanced Data Analytics & Machine Learning

8-Week Advanced Training Program

Comprehensive Session Notes

Week 3 · Day 1

Statistical Rigor & Diagnostics Focus: Logistic Regression for Heart Disease Prediction

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“A model is only as good as its diagnostic checks. Never trust a logistic regression output without validating its underlying assumptions and understanding the prevalence trap.”

About This Document

These comprehensive notes accompany **Week 3, Day 1** of the **Advanced Data Analytics & Machine Learning** program at Mediacrest Training College. Following our foundational work in linear regression and econometric modeling, we now transition into rigorous classification modeling: **Logistic Regression and Advanced Statistical Diagnostics**.

This session bridges classical statistical inference with applied machine learning in a clinical context. We will explore how to handle missingness mechanisms, properly encode features, build and interpret logistic models using Maximum Likelihood Estimation (MLE), and rigorously test assumptions (Linearity of Logits, Multicollinearity, Influential Observations, and Interaction Effects). The session culminates in robust model evaluation using ROC-AUC, Precision-Recall curves, and optimal threshold selection via Youden’s J statistic.

Course at a Glance

	8 Weeks (20+ intensive sessions)
Duration	
Core Tools	Python (pandas, NumPy, statsmodels, scikit-learn, matplotlib, seaborn)
Focus	Inferential statistics, logistic regression, diagnostic testing, clinical risk communication
Capstone	Production-ready ML system with diagnostics, explainability & policy recommendations
Context	Healthcare, Clinical Risk Prediction, and Epidemiological Modeling

Where This Session Fits

Week 3 is titled “**Statistical Rigor, Diagnostics & Advanced Modeling**.”

- Day 1 (this document):** Advanced Data Prep, Logistic Regression, Odds Ratios, Diagnostics (VIF, Cook’s D, Linearity of Logits, Interactions), and Evaluation Metrics.
- Day 2:** Regularization (Ridge/Lasso), Cross-Validation strategies, and handling high-dimensional clinical data.
- Day 3:** Advanced Classification (Random Forests, Gradient Boosting) and model interpretability (SHAP values).

How to Use These Notes

- Follow along in code.** Every listing is runnable. Open the companion Jupyter Notebook and execute each cell as you read.
- Boxes matter.** **Key Concept** boxes hold the theory; **Warning** boxes highlight common pitfalls; **Kenyan Clinical Context** boxes connect material to local healthcare realities; **Mathematical Foundation** boxes provide rigorous statistical proofs.

- **Do the exercises.** Each chapter ends with practice tasks designed to test your statistical intuition.

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1 Advanced Data Preparation for Inference

1.1 Target Audience, Dataset & Dependencies

This module is designed for analysts transitioning to rigorous statistical modeling. We will use the `Heart.csv` dataset, containing clinical cardiovascular data. The required Python dependencies are: `pandas`, `numpy`, `statsmodels`, `scikit-learn`, `matplotlib`, and `seaborn`.

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 import statsmodels.api as sm
6 import statsmodels.formula.api as smf
7 from statsmodels.stats.outliers_influence import variance_inflation_factor
8 from sklearn.model_selection import train_test_split
9 from sklearn.metrics import roc_auc_score, roc_curve, precision_recall_curve,
   auc
10
11 sns.set_theme(style="whitegrid")
12 plt.rcParams['figure.figsize'] = (10, 6)
13
14 # 1. Load Data, explicitly handling 'NA' strings as missing values
15 df = pd.read_csv('Heart.csv', na_values=['NA', '?', 'NaN', ''])
```

Listing 1: Loading the Needed Libraries

1.2 Missingness Mechanisms

Understanding Missing Data

- **MCAR (Missing Completely at Random):** Missingness is unrelated to any observed or unobserved data.
- **MAR (Missing at Random):** Missingness is related to observed data (e.g., older patients might systematically skip the `Ca` test).
- **MNAR (Missing Not at Random):** Missingness is related to the unobserved value itself (e.g., patients with severe disease avoid the `Thal` scan).

Clinical Reality

In clinical data, MAR or MNAR is common. We must impute cautiously to avoid introducing bias into our predictive model.

1.3 Imputation Strategy & Feature Encoding

An initial audit reveals missing values in `Ca` (4 missing) and `Thal` (2 missing).

- **For `Ca` (Ordinal 0-3):** We use median imputation, which is robust to outliers.
- **For `Thal` (Nominal):** We create an explicit `unknown` category to preserve the information that the test was missing, avoiding bias.

```
1 # Imputation
2 df['Ca'] = pd.to_numeric(df['Ca'], errors='coerce')
3 df['Ca'] = df['Ca'].fillna(df['Ca'].median())
4 df['Thal'] = df['Thal'].fillna('unknown')
5
6 # Nominal variables: Use Dummy Encoding (drop_first=True to avoid dummy variable
   trap)
7 df = pd.get_dummies(df, columns=['ChestPain', 'Thal'], drop_first=True, dtype=
   int)
```

Listing 2: Imputation and Dummy Encoding

Pro Tip

Ordinal variables (like Slope, which is already 1, 2, 3, and Ca after imputation) should be retained as numeric. `statsmodels` will handle them appropriately based on the formula specification.

1.4 Checking for Multicollinearity (VIF)

We check core continuous predictors. High Variance Inflation Factor ($VIF > 5$ or > 10) indicates problematic multicollinearity.

```
1 continuous_vars = ['Age', 'RestBP', 'Chol', 'MaxHR', 'Oldpeak']
2 X_vif = df[continuous_vars].copy()
3 X_vif = sm.add_constant(X_vif)
4 vif_data = pd.DataFrame()
5 vif_data["Feature"] = continuous_vars
6 vif_data["VIF"] = [variance_inflation_factor(X_vif.values, i) for i in range(1,
   len(X_vif.columns))]
7 print(vif_data)
```

Listing 3: Variance Inflation Factor (VIF) Check

Output: Max VIF is 1.35 (Age). Since all values are well below 5, **no severe multicollinearity** is detected. No need to drop or combine variables.

2 Building & Interpreting the Logistic Model

2.1 Defining the Statistical Formula

We define the model using `statsmodels` formula API. `C()` denotes categorical variables, and `Treatment('...')` explicitly sets the reference category for interpretation.

```
1 # Drop missing values for this specific model fit demonstration
2 df_clean = df.dropna()
3
4 formula = "HD ~ Age + Sex + C(ChestPain, Treatment('typical')) + RestBP + Chol +
           Fbs + C(RestECG) + MaxHR + ExAng + Oldpeak + C(Slope) + Ca + C(Thal,
           Treatment('normal'))"
```

Listing 4: Model Specification

2.2 Fitting the Logistic Regression Model

We fit the model using Maximum Likelihood Estimation (MLE). Setting `disp=0` suppresses verbose convergence output for cleaner reporting.

```
1 model = smf.logit(formula, data=df_clean).fit(disp=0)
2 print(model.summary())
```

Listing 5: Fitting the Model via MLE

Key Output Interpretation:

- **No. Observations:** 297 (after dropping NAs).
- **Pseudo R-squared (McFadden):** 0.5325. This indicates a strong fit, showing substantial improvement over an intercept-only (null) model.
- **LLR p-value:** 2.124×10^{-36} . The model fits the data significantly better than the null model.

2.3 Interpreting Coefficients and Odds Ratios

From Log-Odds to Odds Ratios

- **Coefficient (β):** Represents the change in the *log-odds* of the outcome for a one-unit change in the predictor.
- **Odds Ratio (OR):** Calculated as $\exp(\beta)$.
 - **OR > 1:** Increased odds of the outcome.
 - **OR < 1:** Decreased odds of the outcome.
 - **OR = 1:** No effect.

```
1 params = model.params
2 conf = model.conf_int()
3 odds_ratios = pd.DataFrame({
4     'Coefficient': params,
5     'Odds Ratio': np.exp(params),
```

```
6     'CI Lower': np.exp(conf[0]),  
7     'CI Upper': np.exp(conf[1]),  
8     'P-value': model.pvalues  
9 })  
10 print(odds_ratios.round(3))
```

Listing 6: Calculating Adjusted Odds Ratios

Top Predictors: Patients with asymptomatic chest pain have 8.06 times the *odds* of heart disease compared to those with typical chest pain (OR = 8.057, 95% CI [2.18, 29.75], $p = 0.002$), holding all else constant.

2.4 CRUCIAL CONCEPT: Odds Ratios (OR) vs. Risk Ratios (RR)

The Prevalence Trap

- **Odds Ratio:** Odds of outcome given exposure / Odds of outcome without exposure. (Native to Logistic Regression).
- **Risk Ratio (Relative Risk):** Probability of outcome in exposed / Probability in unexposed.

If the outcome is rare ($< 10\%$), $OR \approx RR$. However, Heart Disease in this dataset is **~50% prevalent**. Therefore, the OR will **exaggerate** the effect size compared to the RR.

Kenyan Clinical Context

In clinical reporting (e.g., at Kenyatta National Hospital or MP Shah), do not tell a patient they are "8 times more likely" to have heart disease based on an OR. Instead, convert predicted probabilities to marginal risk differences or explicitly state that OR is not RR. Use absolute predicted risks for patient-facing communication.

3 Statistical Diagnostics & Assumptions

3.1 Testing Linearity of Logits

The Assumption

The relationship between continuous predictors and the log-odds of the outcome must be linear.

Practical Diagnostic (Box-Tidwell concept):

1. Bin a continuous variable (e.g., Age into quintiles).
2. Calculate the mean Age and mean HD per bin.
3. Calculate the logit: $\ln\left(\frac{p}{1-p}\right)$ for each bin.
4. Plot Mean Age vs. Logit. A straight line indicates the assumption holds.

```
1 df_clean['Age_Bin'] = pd.qcut(df_clean['Age'], q=5, labels=False)
2 bin_stats = df_clean.groupby('Age_Bin').agg({'Age': 'mean', 'HD': 'mean'})
3 bin_stats['Logit'] = np.log(bin_stats['HD'] / (1 - bin_stats['HD'] + 1e-5))
4 print(bin_stats)
```

Listing 7: Linearity of Logit Check

Result: A regression plot of these points shows a generally linear trend, confirming the assumption holds reasonably well for Age.

3.2 Identifying Influential Observations

Some data points may exert disproportionate leverage on the regression coefficients.

- **Metrics:** Cook's Distance and Leverage (Hat matrix diagonal).
- **Rule of Thumb:** Cook's Distance $> 4/n$ is potentially influential.

```
1 influence = model.get_influence()
2 cooks_d = influence.cooks_distance[0]
3 df_clean['Cooks_D'] = cooks_d
4 n = len(df_clean)
5 threshold = 4 / n
6 influential_obs = df_clean[df_clean['Cooks_D'] > threshold]
7 print(f"Found {len(influential_obs)} potentially influential observations (Cook's D > {threshold:.4f})")
```

Listing 8: Cook's Distance Calculation

Finding: 29 potentially influential observations identified (Threshold ≈ 0.0135).

Best Practice for Outliers

Do not blindly delete outliers. Investigate them first. If they are data entry errors, remove them. If they are genuine extreme clinical cases, consider robust regression techniques or report the model both with and without the influential points.

3.3 Testing for Confounding and Interaction Effects

Question: Does the effect of ExAng (Exercise Induced Angina) on heart disease differ by Sex?

Method: Add an interaction term (Sex:ExAng) to the model.

```
1 interaction_formula = "HD ~ Age + Sex + ExAng + Sex:ExAng"
2 interaction_model = smf.logit(interaction_formula, data=df_clean).fit(dis=0)
3 print(interaction_model.summary().tables[1])
```

Listing 9: Interaction Effect Check

Interpretation: The p-value for Sex:ExAng is 0.238 (> 0.05). There is **insufficient evidence** that the effect of Exercise Induced Angina on heart disease differs significantly according to Sex (no effect modification).

4 Evaluation & Thresholding

4.1 Train/Test Split Strategy

To avoid overly optimistic in-sample metrics and assess generalizability, we split the data.

```
1 X = df_clean.drop(columns=['HD', 'Age_Bin', 'Cooks_D', 'Leverage'])
2 y = df_clean['HD']
3 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
    random_state=42, stratify=y)
```

Listing 10: Train/Test Split

Note on Demonstration

For the simplicity of visualizing curve shapes in this script, full dataset probabilities are used. In strict production, predictions must be mapped exclusively through `X_test`.

4.2 ROC-AUC and Precision-Recall Curves

- **ROC-AUC (0.935):** Plots True Positive Rate (Sensitivity) vs. False Positive Rate (1 - Specificity). An AUC of 0.935 indicates an excellent ability to distinguish between patients with and without heart disease.
- **Precision-Recall AUC (0.929):** Crucial for high-stakes clinical data where False Positives (unnecessary angiograms) and False Negatives (missed diagnoses) carry heavy, asymmetric costs. It demonstrates strong predictive performance in balancing Precision (Positive Predictive Value) and Recall (Sensitivity).

4.3 Threshold Selection for Clinical Policy

The default classification threshold is 0.5, which may not be optimal for clinical trade-offs. We optimize using **Youden's J statistic**.

Youden's J Statistic

$$J = \text{Sensitivity} + \text{Specificity} - 1 \quad (1)$$

The threshold that maximizes J is selected as the optimal cut-off.

```
1 fpr, tpr, roc_thresholds = roc_curve(y, model.predict(df_clean))
2 youden_j = tpr - fpr
3 optimal_idx = np.argmax(youden_j)
4 optimal_threshold = roc_thresholds[optimal_idx]
5 print(f"Optimal Classification Threshold: {optimal_threshold:.3f}")
```

Listing 11: Optimal Threshold Selection

Result: Optimal Classification Threshold = **0.481**. At this threshold, we optimally balance False Positives and False Negatives for clinical policy.

5 Best Practices, Checklist & Deliverables

5.1 Professional Preprocessing & Diagnostics Checklist

mclightStage	Action	Rationale
Missingness	Audit & Impute (Median / Unknown).	Prevents bias from MCAR/MAR/MNAR mechanisms.
Encoding	Dummy encoding with drop_first=True.	Avoids the dummy variable trap (perfect multicollinearity).
Multicollinearity	Check VIF < 5.	Ensures model coefficients are stable and interpretable.
Linearity	Bin continuous vars & plot Logit.	Validates the core logistic regression assumption.
Influential Points	Cook's Distance > 4/n.	Identifies points that disproportionately dictate the model.
Interactions	Test biologically plausible interactions.	Checks for effect modification (e.g., Sex × ExAng).
Evaluation	ROC-AUC, PR-AUC, Youden's J.	Assesses discrimination and sets optimal clinical thresholds.

Table 1: The Professional Logistic Regression Diagnostics Checklist.

5.2 Session Deliverable: The Statistical Diagnostics Report

The final output of this pipeline is a structured, reproducible report. Below is the synthesized deliverable generated by the script.

STATISTICAL DIAGNOSTICS REPORT: Heart Disease Prediction
1. DATA PREPARATION - Missingness handled: Ca imputed with median (0.0); Thal imputed as unknown. - Encoding: Dummy variables applied to nominal features; ordinal features retained. 2. MULTICOLLINEARITY (VIF) - Max VIF observed: 1.35 (Age). - Recommendation: No severe multicollinearity detected. Model coefficients are stable. 3. MODEL FIT & ODDS RATIOS - Pseudo R-squared (McFadden): 0.533 - Top 3 Predictors by Odds Ratio: • ChestPain (asymptomatic): OR = 8.06, <i>p</i> = 0.002 • Sex: OR = 4.69, <i>p</i> = 0.004 • Thal (reversible): OR = 4.03, <i>p</i> = 0.001 4. DIAGNOSTICS - Linearity of Logit: Visually inspected and holds. - Influential Observations: 29 records exceed Cook's D threshold of 0.0135 (warrants review).

- Interaction Check: **ExAng * Sex** interaction p-value = 0.2375 (not significant).

5. EVALUATION METRICS

- ROC-AUC: 0.935
- Precision-Recall AUC: 0.929
- Recommended Policy Threshold: 0.481

NOTE ON CLINICAL REPORTING:

Odds Ratios (OR) are reported above. Because the prevalence of Heart Disease in this cohort is high (~50%), the OR will **overstate** the Risk Ratio (RR). Do not interpret OR as "X times more likely"; instead, use marginal predicted probabilities for patient-facing risk communication.

Final Deliverable: The Clinical Memo

Draft a 1-page technical memo addressed to the Chief Medical Officer. Your memo must include:

1. **Model Justification:** Briefly explain why logistic regression was chosen and how missingness was handled to preserve clinical integrity.
2. **Diagnostic Validation:** Report the VIF, Cook's Distance findings, and the interaction check, explicitly stating whether the model is stable.
3. **Clinical Interpretation:** Translate the top Odds Ratios into plain-English risk assessments, **explicitly avoiding** the "X times more likely" fallacy, and recommend the 0.481 threshold for triaging patients for further cardiac testing.

6 Glossary of Key Terms

- Cook's Distance** A measure of the influence of a single observation on the fitted regression coefficients. Values $> 4/n$ warrant investigation.
- Linearity of Logits** The assumption that the relationship between continuous predictors and the log-odds of the outcome is linear.
- Maximum Likelihood Estimation (MLE)** The method used to estimate the parameters of a logistic regression model by maximizing the likelihood of observing the given data.
- Odds Ratio (OR)** The ratio of the odds of an event occurring in one group to the odds of it occurring in another group.
- Precision-Recall AUC** The area under the Precision-Recall curve, highly informative for evaluating models on imbalanced datasets or where false positives/negatives have asymmetric costs.
- Risk Ratio (RR)** The ratio of the probability of an outcome in an exposed group to the probability in an unexposed group. Preferred over OR for clinical communication when prevalence is high.
- Youden's J Statistic** A single statistic that captures the performance of a dichotomous diagnostic test, defined as $J = \text{Sensitivity} + \text{Specificity} - 1$. Used to find the optimal classification threshold.

7 References & Further Reading

- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.
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